Positive coefficients for the elliptic-integral upper remainder and its companion functions

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Task 1: Resolve MathDB 350004. We complete Conjecture 5.1 of Qiu–Bao–Ma–Jiang [1]. The principal new step is positivity of every coefficient of the upper remainder h_{11}. For the quotient G, we explicitly use the recently published theorem of Yang–Zhao [2]; the resulting proof does not claim that already established part as new.

Put $x = r^2$, and write

$$F_0(x) = {}_2F_1(\tfrac{1}{2}, \tfrac{1}{2}; 1; x) = \frac{2}{\pi}\mathcal{K}(\sqrt{x}), \qquad F_1(x) = {}_2F_1(\tfrac{1}{2}, 1; \tfrac{3}{2}; x) = \frac{\operatorname{arctanh}\sqrt{x}}{\sqrt{x}}.$$

Let $A = \log F_0$, $L = \log F_1$, and $Q = A/L$, with the removable value $Q(0) = 3/4$. The six functions in the conjecture are

$$G(r) = Q(x), \qquad f(r) = \frac{Q(x) - 3/4}{x}, \qquad f_7(r) = \frac{L(x)}{x},$$
$$h_{11}(r) = \frac{H(x)}{x^2}, \qquad H(x) = \frac{3+x}{4}L(x) - A(x), \qquad h_{13}(r) = \frac{A(x)}{x},$$
$$h_{12}(r) = \frac{A(x) - (3/4 + x/320)L(x)}{x^3}.$$

Theorem 1. *Each of $f, G, f_7, h_{11}, h_{12}, h_{13}$ has a convergent expansion $\sum_{n\ge0} c_n r^{2n}$ for $0 \le r < 1$, with $c_n > 0$ for every n. In particular all six functions are strictly absolutely monotone on $(0,1)$: every derivative, including order zero, is positive there. The function f is strictly convex, and h_{11}, h_{12} are strictly increasing and strictly convex, with ranges*

$$h_{11}((0,1)) = (79/960, \log(\pi/2)), \qquad h_{12}((0,1)) = (517/604800, \infty).$$

As the displayed source expansions make clear, positive coefficients refers to the even-power series. Odd powers have zero coefficients, since all six functions are even.

A coefficient-sign lemma. We give the elementary reciprocal argument underlying the proof, so no unproved coefficient-positivity criterion is needed.

Lemma 1. *Suppose $B(x) = \sum_{n\ge0} b_n x^n$, $b_0 = 1$, $b_n > 0$, has radius of convergence at least one, and b_n/b_{n-1} is nondecreasing. Then*

$$\frac{1}{B(x)} = 1 - \sum_{n\ge1} c_n x^n, \qquad 0 \le c_n \le b_n.$$

If the ratios are strictly increasing, every c_n is positive. If $U(x) = \sum_{n\ge0} u_n x^n$ has positive coefficients and u_n/b_n is nonincreasing, every nonconstant coefficient of U/B is nonpositive. It is strictly negative when both ratio conditions are strict.

Proof. Coefficient comparison determines

$$c_n = b_n - \sum_{j=1}^{n-1} c_j b_{n-j}.$$

Inductively, $c_1 = b_1 > 0$, and

$$\sum_{j=1}^{n-1} c_j b_{n-j} \le \frac{b_n}{b_{n-1}} \sum_{j=1}^{n-1} c_j b_{n-1-j} = b_n.$$

Thus $0 \le c_n \le b_n$. For strictly increasing ratios the inequality is strict when $n \ge 2$, already in the $j = 1$ term. The bound $c_n \le b_n$ ensures convergence on the unit disk. For $n \ge 1$, set $\rho_n = u_n/b_n$. Then

$$[x^n]\frac{U}{B} = u_n - \sum_{j=1}^{n} c_j u_{n-j} = \sum_{j=1}^{n} c_j b_{n-j}(\rho_n - \rho_{n-j}) \le 0.$$

Strictness follows from the $j = n$ term. □

Three quotients with negative coefficients. Define

$$G_0(x) = {}_2F_1(\tfrac{1}{2}, \tfrac{1}{2}; 2; x), \qquad G_1(x) = {}_2F_1(\tfrac{1}{2}, 1; \tfrac{5}{2}; x).$$

Direct series comparison gives

$$F_0' = \frac{G_0}{4(1-x)}, \qquad F_1' = \frac{G_1}{3(1-x)}. \tag{1}$$

Write $a_n = ((1/2)_n/n!)^2$. The four coefficient sequences are

$$[x^n]F_0 = a_n, \quad [x^n]F_1 = \frac{1}{2n+1}, \quad [x^n]G_0 = \frac{a_n}{n+1}, \quad [x^n]G_1 = \frac{3}{(2n+1)(2n+3)}.$$

The denominator sequences of G_0/F_0, G_1/F_1, and G_0/G_1 are strictly log-convex: their consecutive ratios are respectively

$$\left(\frac{n-1/2}{n}\right)^2, \qquad \frac{2n-1}{2n+1}, \qquad \frac{2n-1}{2n+3} \qquad (n \ge 1),$$

all strictly increasing. For the first two quotients, the ratios of numerator to denominator coefficients are $1/(n+1)$ and $3/(2n+3)$, both strictly decreasing. For the third, if $\rho_n = [x^n]G_0/[x^n]G_1$, then

$$\frac{\rho_{n+1}}{\rho_n} = \frac{(n+1/2)(n+5/2)}{(n+1)(n+2)} < 1.$$

Lemma 1 therefore proves that all three quotients have constant coefficient one and strictly negative coefficients at every positive degree.

The upper remainder has positive coefficients. First, Lemma 1 gives $1/F_1 = 1 - C$, where every coefficient of C, starting at degree one, is positive. For $0 < x < 1$, $C(x) < 1$. Thus

$$L(x) = -\log(1 - C(x)) = \sum_{j \geq 1} \frac{C(x)^j}{j}$$

has strictly positive coefficients at all positive degrees. The same argument applies to $A = \log F_0$.

Next set

$$P_0(x) = \frac{(1-x)^{-1/4}}{F_0(x)}, \qquad P_1(x) = \frac{(1-x)^{-1/3}}{F_1(x)}.$$

By (1),

$$(\log P_0)' = \frac{1 - G_0/F_0}{4(1-x)}, \qquad (\log P_1)' = \frac{1 - G_1/F_1}{3(1-x)}.$$

Both right sides have nonnegative coefficients. Integration from zero and exponentiation show that P_0, P_1 have nonnegative coefficients and constant coefficient one. Consequently so does

$$\frac{1}{(1-x)F_0F_1} = (1-x)^{-5/12} P_0 P_1. \tag{2}$$

Introduce the source's auxiliary function

$$J(x) = F_0(x) - F_1(x)\frac{F_0'(x)}{F_1'(x)} = F_0(x) - \frac{3}{4}F_1(x)\frac{G_0(x)}{G_1(x)}.$$

Differentiation cancels the first-order terms:

$$J'(x) = -\frac{3}{4}F_1(x)\left(\frac{G_0(x)}{G_1(x)}\right)'.$$

The preceding quotient-sign result proves that every coefficient of J' is strictly positive, and $J(0) = 1/4$. Since $a_n < a_{n-1}$ for $n \geq 1$, the series $(1-x)F_0$ has constant coefficient one and negative coefficients thereafter. It follows that

$$T(x) = 4J(x) - (1-x)F_0(x)$$

has zero constant coefficient and strictly positive coefficients at every positive degree.

The following identity now closes the proof for h_{11}:

$$4H'(x) = L(x) + \frac{G_1(x)T(x)}{3(1-x)F_0(x)F_1(x)}. \tag{3}$$

It follows directly by differentiating H and using (1). Every factor on the right has nonnegative coefficients, by (2); L supplies a strictly positive coefficient at every positive degree. The constant coefficient is zero. As $H(0) = 0$, integration proves

$$H(x) = \sum_{n \geq 2} d_n x^n, \qquad d_n > 0.$$

Dividing by x^2 proves precisely the required coefficient positivity of h_{11}.

The other five functions. Yang–Zhao's Theorem 1.1 and its coefficient conclusion (6.11)–(6.15) [2] establish

$$Q(x)=\sum_{n\ge 0} q_n x^n, \quad q_n>0, \quad q_0=\frac{3}{4}, \quad q_1=\frac{1}{320}, \quad q_2=\frac{517}{201600}, \qquad 0\le x<1. \tag{4}$$

We use this established result with attribution. It immediately proves the positive coefficients of G and $f=(Q-q_0)/x$. The previous logarithm argument proves those of $f_7=L/x$ and $h_{13}=A/x$. Finally,

$$h_{12}(r)=\frac{L(x)}{x}\,\frac{Q(x)-q_0-q_1x}{x^2}$$

is a product of two series having strictly positive coefficients. This proves every coefficient assertion.

All series converge on the unit disk. This follows from the coefficient bounds in Lemma 1, the displayed convergent products and logarithms, and (4). Termwise differentiation is therefore valid locally on $(0,1)$. For any derivative order in r, infinitely many even monomials of sufficiently high degree give positive contributions. This proves strict absolute monotonicity, strict increase, and strict convexity as stated.

Endpoint values and ranges. The first two logarithm coefficients are

$$A(x)=\frac{x}{4}+\frac{7x^2}{64}+O(x^3), \qquad L(x)=\frac{x}{3}+\frac{13x^2}{90}+O(x^3).$$

Thus $H(x)=79x^2/960+O(x^3)$; also (4) gives

$$h_{12}(0+)=\frac{1}{3}q_2=\frac{517}{604800}.$$

As $x\uparrow 1$, the standard elementary/elliptic asymptotics give

$$F_0(x)\sim\frac{1}{\pi}\log\frac{1}{1-x}, \qquad F_1(x)\sim\frac{1}{2}\log\frac{1}{1-x}.$$

Consequently $L-A\to\log(\pi/2)$, $L\to\infty$, and $(1-x)L\to 0$. Therefore

$$H=L-A-\frac{1-x}{4}L\longrightarrow\log(\pi/2),$$

while

$$A-(3/4+x/320)L=\left(1-\frac{3}{4}-\frac{x}{320}\right)L+(A-L)\longrightarrow+\infty.$$

Strict increase and continuity give both exact open ranges. Theorem 1, and hence the entire conjecture, follows.

Task 2: Extend the upper-remainder argument. Determine the hypergeometric parameter pairs for which an analogue of $H(x)=((3+x)/4)\log F_1(x)-\log F_0(x)$ has positive coefficients after removing its initial zeros.

Task 3: Optimize bounds from truncated remainders. Use the positive coefficients of h_{11} and h_{12}, together with their endpoint behavior, to derive sharp two-sided bounds for $\mathcal{K}(r)$ at any prescribed truncation degree.

Task 4: Describe the upper-remainder coefficients. Determine asymptotic formulas and any Hausdorff-moment or log-convexity properties of the sequence $(d_{n+2})_{n\geq 0}$.

Task 5: Find a direct proof for the quotient coefficients. Seek an elementary coefficient argument for (4) that integrates with Lemma 1 and extends to nearby hypergeometric parameter families.

References

[1] S.-L. Qiu, Q. Bao, X.-Y. Ma, and H.-B. Jiang, On a Conjecture Concerning the Approximates of Complete Elliptic Integral of the First Kind by Inverse Hyperbolic Tangent, arXiv:2103.04072v1 (2021), Conjecture 5.1, p. 21; Lemma 2.2 and equation (4.2).

[2] Z.-H. Yang and T.-H. Zhao, On the absolute monotonicity related to Alzer–Qiu's conjecture, *Integral Transforms and Special Functions*, published online 13 April 2026, doi:10.1080/10652469.2026.2657542, Theorem 1.1 and equations (6.11)–(6.15).